

Payments, Processors & IT Services

The Long View: Payments - Does Cash Digitization Runway Still Exist?



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We publish the 12th annual edition of our global payments forecast model. A lot has changed in the last 10+ years - U.S. is now highly penetrated by cards (72% vs. 58% ten years ago), local payment methods have gained traction (e.g., in Asia where card volume growth has somewhat stalled), domestic schemes have lost share (e.g., in Europe), other alternative ways to pay (e.g., A2A) have generated noise but have remained contained, and stablecoins are finding some utility in markets where dollars are desirable. Last year was the first time when we brought our forecasts down (see [link](#)) - largely because of market maturing in the U.S. and regional dynamics in Asia. Against that dynamic backdrop, our global payments forecast model remains the under-pinning of our investment view on the sector and V/Ma in particular - even as value-added services capture a growing share of revenue.

We forecast 8% cc (~7-8% nominal) card volume growth from 2025-30E (below 2022-25 CAGR of 9% cc), driven by 5-6% PCE growth and 2-3% card penetration-driven growth (penetration growing to 71%, up from 64% in 2025). The exit growth rate in 2030 in our model is 7%. **We forecast deceleration vs. history (~8% cc vs. 9% cc historically)** due to already high penetration in the U.S. (where we now forecast ~5-6% card volume growth CAGR vs. ~6% CAGR between 2022-25), slowdown in Asia (due to local market nuances). Europe remains an attractive market for cards: we forecast 11% cc volume growth driven by continental Europe (e.g., Germany, Italy, Spain) vs. 16% historically. We are monitoring *optionality* driven by possible inflection in eCommerce from Agentic commerce and autonomous agent transactions as a growth driver for card payments but note that it's simply too early to bake anything into our numbers. We will closely monitor how cards' commercial models to address historically tougher to address categories such as rent and healthcare.

We forecast a stronger growth at 9% in the number of transactions globally between 2025-30E (10% 2022-25). Card penetration is lower when assessed as a % of number of transactions (as opposed to transaction value). V/Ma derive >1/3rd of their gross revenue from # of transactions, which tend to grow faster than \$ volumes. Cross-border is at an interesting point, and we will address that in a separate note.

Card volumes only tell part of the story. **We see a runway in tokens, VAS penetration and optionality in new flows** (more details in a separate note), **which along with 9% growth in transactions can drive DD% revenue growth** despite card volume growth of 8%. For reference, even in high card penetration markets, networks have historically demonstrated an ability to grow revenues at double digits. Domestic schemes (~\$4T of volumes globally) are an underappreciated opportunity e.g., Visa gained 5-6ppt share in Continental Europe vs Girocard / CB card in Germany/France in 2024.

Alternative ways-to-pay drive HSD-LDD% of consumer-to-business payments globally. While account-to-account and local methods are a risk to debit (especially in certain markets), even where A2A/local payments have had success (e.g., India, Brazil), *credit-card* volumes are growing HSD-LDD%. Finally, on a 5yr time horizon, **we view stablecoins as a solution looking for a problem** in retail consumer payments ([note](#)).

INVESTMENT IMPLICATIONS

We rate V, MA, ADYEN, XYZ and TOST OP. We rate PYPL, FIS, FISV, GPN, KLAR MP.

PLEASE SEE LINKS TO OUR RECENT RESEARCH

- [Weekend Tech Byte: The 'cost' of cards — the often misleading comparisons; our perspectives in key charts](#) (June 2026)
- [Visa: Reflections on our Fireside Chat with CEO Ryan McInerney](#) (June 2026)
- [Payments: Stablecoins — What we learned from our dozens of industry conversations; Slide deck](#) (May 2026)
- [Visa, Mastercard: Numbers look good but will it matter?](#) (April 2026)
- [Visa, Mastercard: The bull case; slide deck from deep dive call on Agentic, Stablecoins, VAS and Regulation](#) (March 2026)
- [The Age of Agents: Insights from our Inaugural Agentic Commerce Day](#) (April 2026)
- [Payments: Mastercard's BVNK acquisition; what Stablecoins truly mean for payments](#) (March 2026)
- [Visa, Mastercard: Payments-ocalypse, will AI doom the networks?](#) (February 2026)
- [Payments: The growing regulatory risk premium? But will the numbers be impacted in the medium-term?](#)
- [Payments/Fintech 2026: A year to remember? Top themes and stock ideas](#) (January 2026)
- [Visa, Mastercard: Why we like the networks more now vs. 5yrs ago](#) (October 2025)
- [Stablecoins: Let's Play Out a Scenario for Disruption in Consumer Payments..](#) (June 2025)
- [Payments: What happens in a recession?](#) (Mar 2025)

DETAILS

OUR FORECAST: ~8% CC (~7-8% NOMINAL) CARD VOLUME GROWTH FROM 2025-30

We forecast ~8% constant currency (~7-8% nominal) C2B purchase volume CAGR over the next 5 years - slightly below the ~9% growth between 2022-25 and ~11% between 2019-22 ([Exhibit 3](#)).

Card penetration grew 2-3ppt/year from 2022-25. We estimate it to grow ~1-2ppt/year going forward ([Exhibit 1](#)).

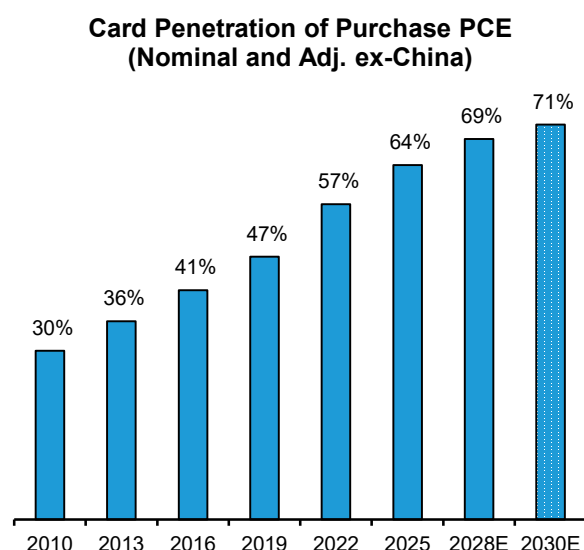
Card penetration grew ~1ppt in 2025. The ~1ppt penetration growth in 2025 was driven by healthy penetration growth in regions such as Europe (up 3ppt), LATAM (3ppt), MEA (~3ppt), offset by more muted growth in Canada (90%+ penetration), US (flat) and APAC (~1ppt; flat ex-V/MA volumes in China). The pandemic very clearly accelerated cash digitization through faster growth in eCommerce, an increase in contactless usage, greater adoption of digital payments by merchants and by bringing in new digital payment users. From 2019-22, global card penetration grew ~3ppt/year, twice the historical average rate.

We forecast deceleration vs. history (~8% cc vs. 9%) due to already high penetration in the U.S. (where we forecast ~5-6% card volume growth CAGR vs. ~6% CAGR between 2022-25), slowdown in Asia (due to local market nuances in certain markets and unique dynamics in mainland China - see details in the note). **Our estimates embed the fact that many developed markets are now at high card penetration and EM is a battleground.** In 2025, we estimate US card penetration at ~72%, up from ~67% in 2019. We estimate markets representing ~50% of global GDP (e.g., US, UK, Canada, Australia, South Korea) are now already at >80% digital payments penetration.

Note:

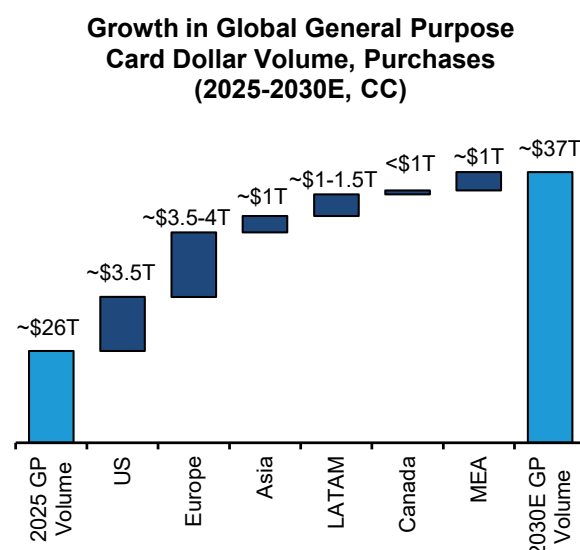
- Our forecasts do not include meaningful tailwind from new payment flows, nor do they incorporate benefits from agentic commerce (it's simply too early to bake into our numbers). We also exclude Visa direct (in addition to commercial cards) from the numerator. Our forecast doesn't fully incorporate new ways to pay i.e. cards used to fund digital wallets as this can be quite tricky for a few markets.
- Our card penetration numbers might differ slightly vs our prior estimates as we update our denominator (i.e. PCE numbers) to reflect recent data.

EXHIBIT 1: We expect card penetration to approach ~71% by 2030, up from 64% in 2025



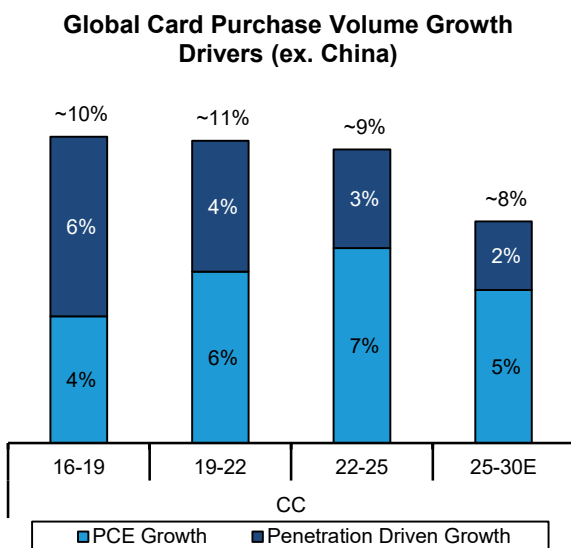
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 2: Majority of card volume growth from 2025 to 2030E will still be driven by US and Europe



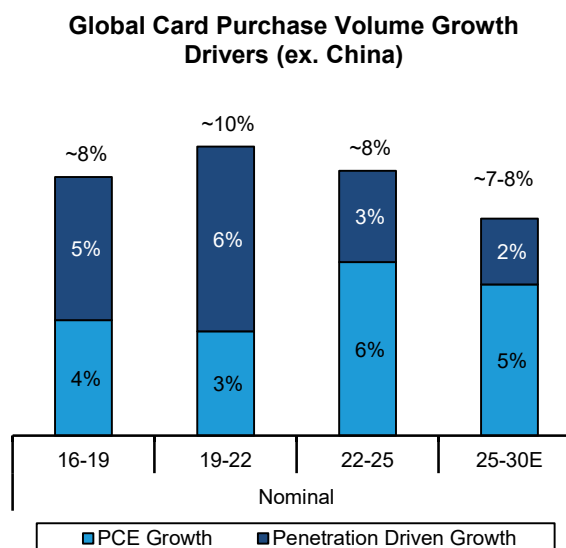
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 3: On a cc basis, card volumes grew at ~11% between 2019-22, ~9% between 2022-25 and we forecast ~8% growth between 2025-30.



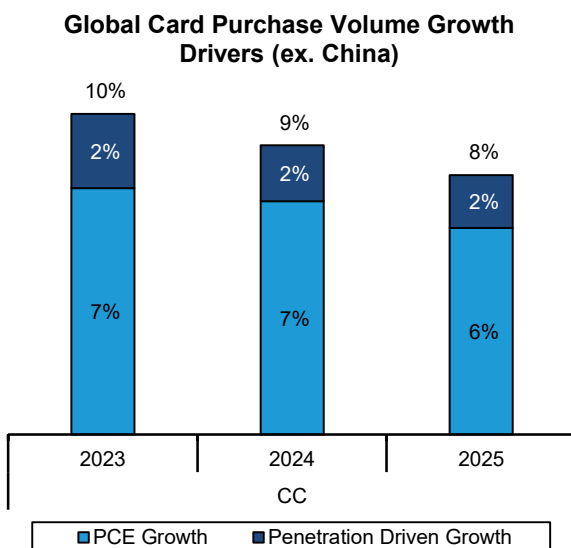
Note: Exclusion of Russia has a modest negative impact on 2019-22, 22-25 growth CAGRs
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 4: On a nominal basis, we forecast ~7-8% card volume growth over 2025-30E globally.



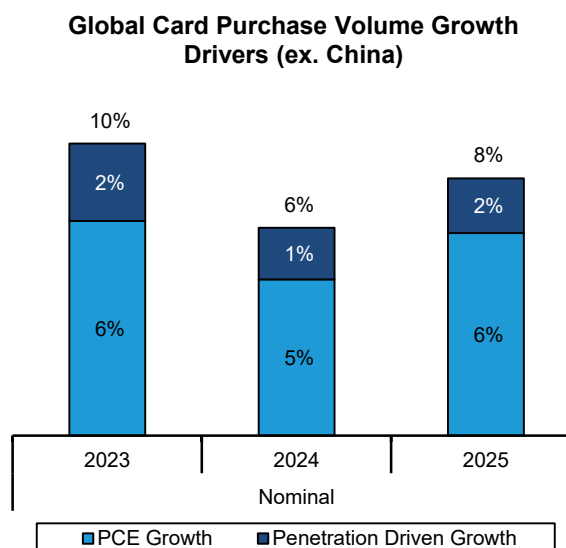
Note: Exclusion of Russia has a modest negative impact on 2019-22, 22-25 growth CAGRs
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 5: Card volumes grew at ~8% on a nominal basis in 2025..



Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 6: ..and also on cc basis



Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

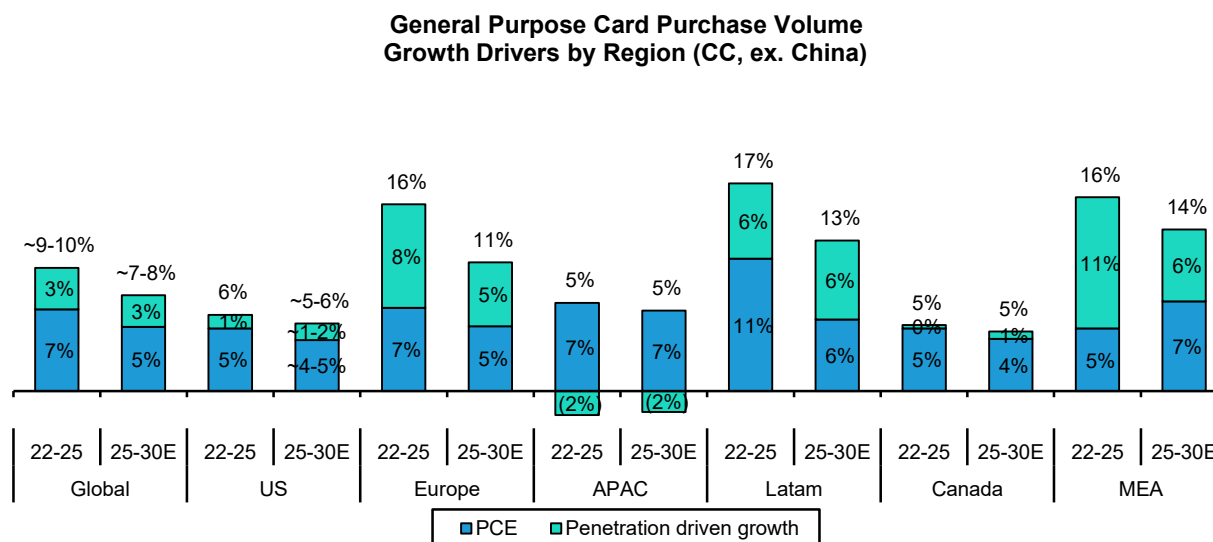
CARD VOLUME GROWTH WILL LIKELY BE MORE CYCLICAL VS. HISTORY AS PENETRATION-DRIVEN GROWTH SOMEWHAT EASES

On a nominal basis, our ~7-8% volume CAGR forecast from 2025-30E comprises ~5% underlying global PCE growth & ~2ppt of secular cash to card conversion growth.

Our 5-yr nominal growth outlook by region is ~5-6% for U.S., ~11% cc (~10% nominal) for Europe, ~5% cc (~4% nominal) for Asia, ~13% for Latam cc (similar on nominal basis), 5% for Canada cc (~6% nominal) and 14% cc (12% nominal) for MEA. We estimate that U.S. is ~72% penetrated by cards (up from ~67% in 2019), Europe at ~77% (up from 44% in 2019), Asia-Pac (ex-mainland China) at ~41% (up from 36% in 2019), Latam at ~48% (up from 29%), Canada at ~92% (up from 79% in 2019) and MEA at ~61% (up from 26%) ([Exhibit 9](#)).

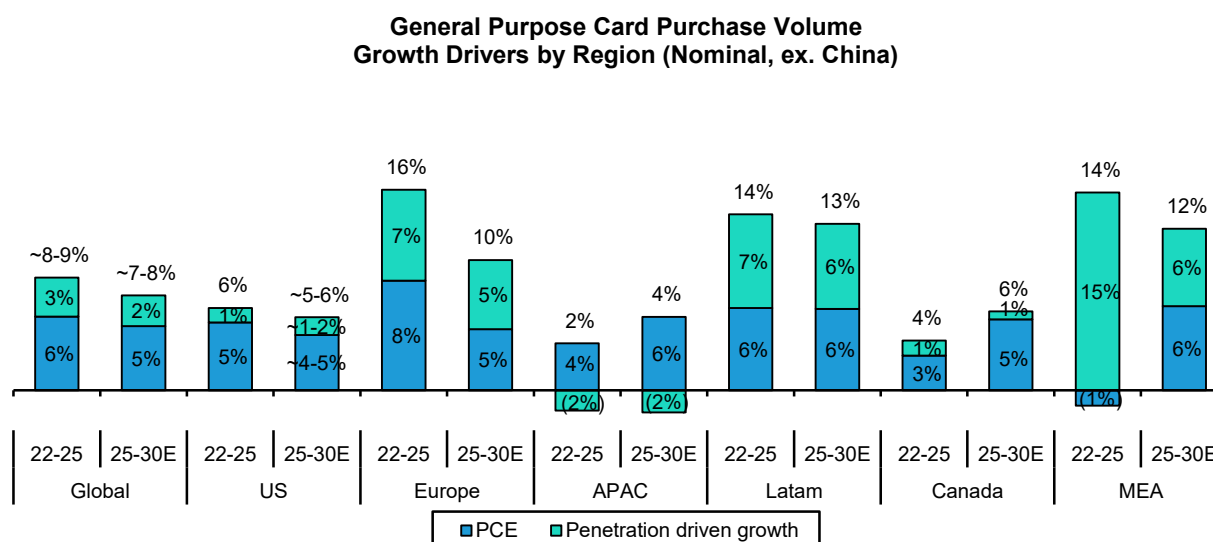
We currently exclude mainland China from our forecasts but note that Mastercard went live in mainland China in May 2024 after acquiring a domestic license. Further, in APAC, in addition to mainland China, there are also certain local market dynamics which we will discuss later in the note.

EXHIBIT 7: Our growth forecast by region on cc basis - US - ~5-6%, Europe - 11%, APAC - 5%, LATAM - 13%, Canada - 5%, MEA - 14%

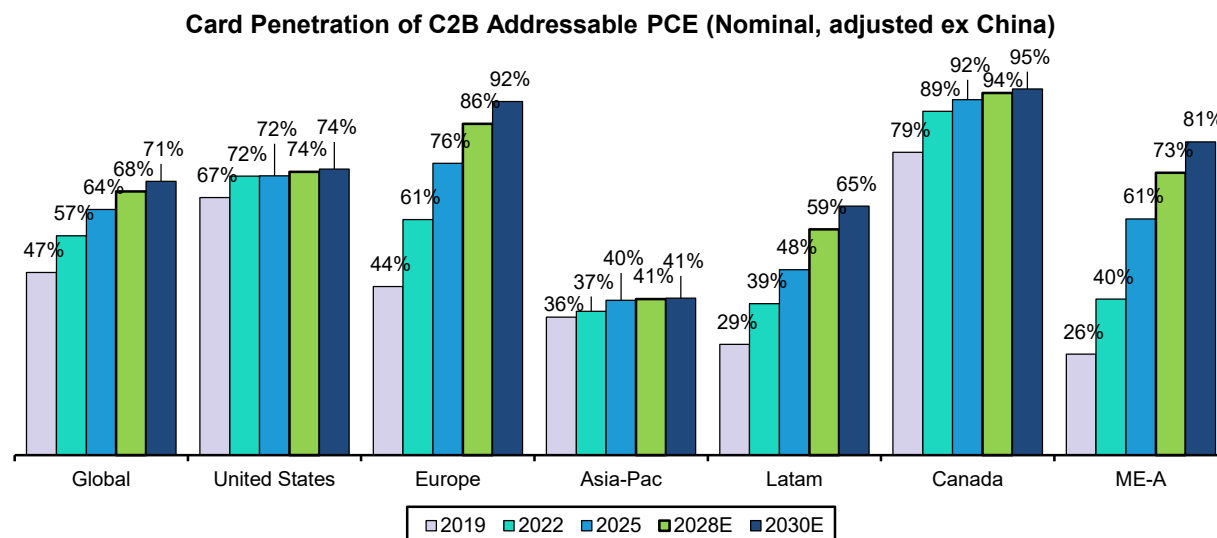


Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 8: Our nominal growth forecast by region - US - ~5-6%, Europe - 10%, APAC - 4%, LATAM - 13%, Canada - 6%, MEA - 12%



Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 9: **We expect card penetration growth to be driven by Europe, Latam, and MEA**

For APAC, the numerator excludes V/MA from China from General Purpose Card Volumes in the previous exhibits

Source: Nilson, World Bank, IMF and Bernstein estimates and analysis

THERE IS STILL A FAIR AMOUNT OF CASH/CHECK THAT EXISTS GLOBALLY

Although it sure feels like cards are everywhere, **there is still a good amount of cash/ check in existence globally.**

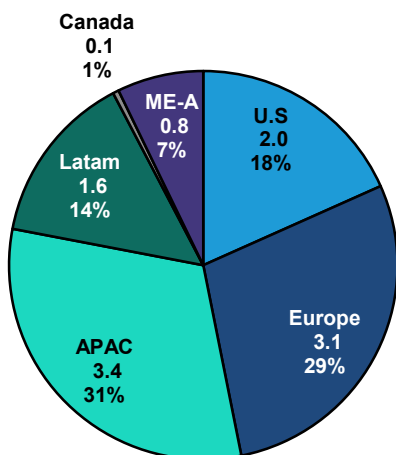
Consumer-to-business (C2B) payments is a \$40T market and is currently 64% penetrated by cards (vs. 47% in 2019).

We expect card penetration to approach ~70-75% by 2030. We est. ~\$11T of cash/check opportunity globally split across different regions e.g., U.S./Canada (~\$2T), Europe (~\$3T), Asia-Pac ex-mainland China (~\$3-4T), Latam (~\$1.5-2T), MEA (~\$0.5-T) in 2025 ([Exhibit 10](#)). We estimate that the top 25 markets by GDP combined have at least ~\$3-4T of cash (excluding checks).

Catalysts for continued cash digitization — contactless, e-commerce and possibly agentic commerce. Contactless has proven to be a powerful catalyst for accelerating cash-to-card conversion. And these payments are a great tool for habit formation in digital purchases through very sticky every day use-cases such as transit, groceries, vending machines, etc. Global tap-to-pay or contactless penetration of face-to-face transactions for MA grew 5ppt in 2025 to 77% (78% in 1Q26) and, up from 45% in 2021. eCommerce and the era of agentic transactions, potentially micropayments are also catalysts that can accelerate volume and transactions (more on this below) growth.

EXHIBIT 10: **Although it sure feels like cards are everywhere, there is still a surprising amount of cash/check in existence globally**

**Global Cash/Other volumes by region
(2025E, Total \$11T)**



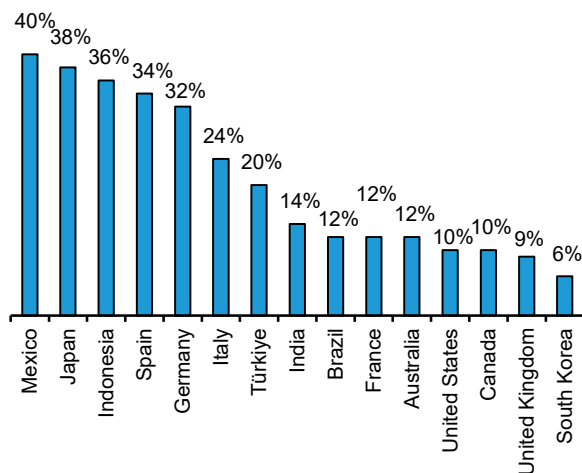
Source: Nilson, WEO, World Bank, Mastercard Whitepaper (Cashless journey), Corporate reports and Bernstein estimates and analysis

DIGITAL PAYMENTS PENETRATION VARY SIGNIFICANTLY BY COUNTRY

Digital payments (and cards) penetration vary significantly by country **and payments are local and how people pay truly varies by country**. Cards being one of the most popular payment methods, is often complemented with digital wallets, RTP networks, etc. depending on the country you look at.

EXHIBIT 11: **Cash Usage varies widely by country; Mexico, Japan, Indonesia, Germany still have high amounts of cash..**

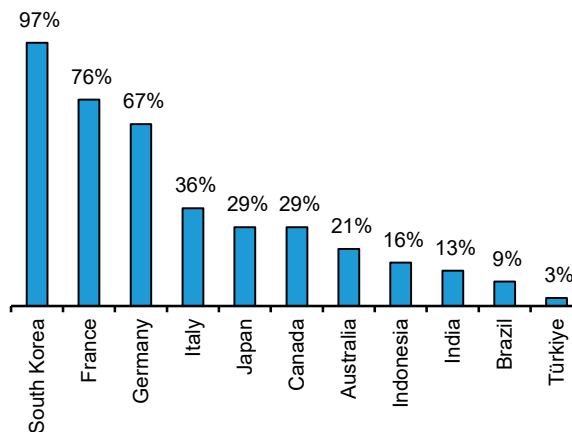
**Cash as a % of POS Transaction Value
(Top 15 GDP Countries, 2025)**



Source: Global Payments Report, Bernstein analysis

EXHIBIT 12: **Share of domestic schemes also varies widely by country..**

**Share of Domestic Schemes within
Overall Network (% , 2024)**



South Korea includes BC , Hana, Hyundai, KB Kookmin, Lotte,

NH, Samsung, Shinhan, Woori; France includes Cartes Bancaires; Germany includes Girocard; Italy includes Poste Italiane, Bancomat; Japan includes JCB;

Canada includes Interac; Australia includes EFTPOS; Indonesia includes GPN;
 India includes Rupay; Brazil includes Elo; Turkey includes Troy
 Source: Global Payments Report, Bernstein analysis

EXHIBIT 13: How consumers 'pay' around the world



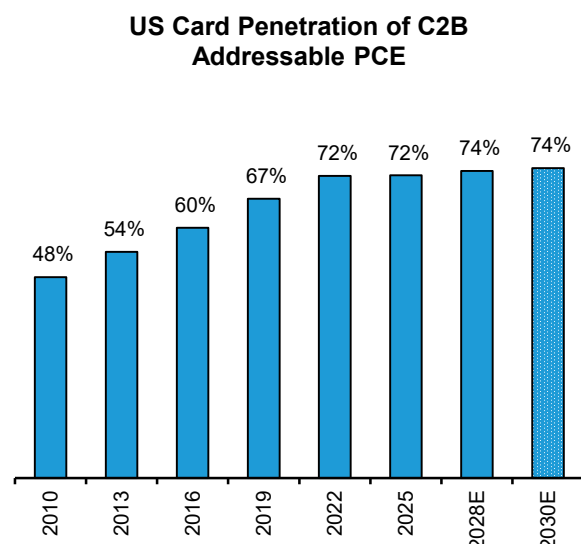
Source: Global Payments Report, Company reports, Bernstein analysis

HOW PENETRATED IS THE US (FOR CARDS)?

We estimate US 'consumer' card penetration at 74% by 2030. We scrub the numbers a fair bit. Our numerator includes US card volumes (reported by V, MA, AXP and DFS) plus Private Label and PIN debit, and minus Commercial Cards and Visa Direct. Our definition of addressable PCE = PCE — Imputed Rentals (essentially mortgages) — Healthcare (which is not out-of-pocket) — Certain Financial Services.

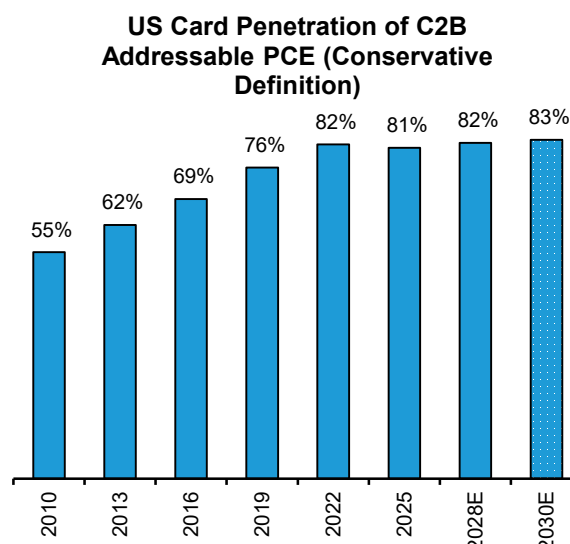
If, however, we use a more conservative definition of PCE (i.e. exclude rents, education and bill payments), we get to 83% card penetration. For the more 'conservative' PCE definition, we also deduct housing (actual rentals), education and utilities payments. Overall, we forecast ~5-6% US card volume growth between 2025-30— vs. ~6% from 2022-25 and ~13% from 2019-22 on lower penetration-driven growth.

Have US payments reached maturity? The US 'consumer' card payments market has matured, regardless of the definition we use for addressable PCE in order to assess the market. Between 2022-25, card penetration in the US has not grown, remaining somewhat stable at 72%. We are monitoring *optionality* driven by possible inflection in eCommerce from Agentic commerce and autonomous agent transactions as a growth driver for card payments but note that it's simply too early to bake anything into our numbers.

EXHIBIT 14: We estimate US 'consumer' card penetration at 72% in 2025

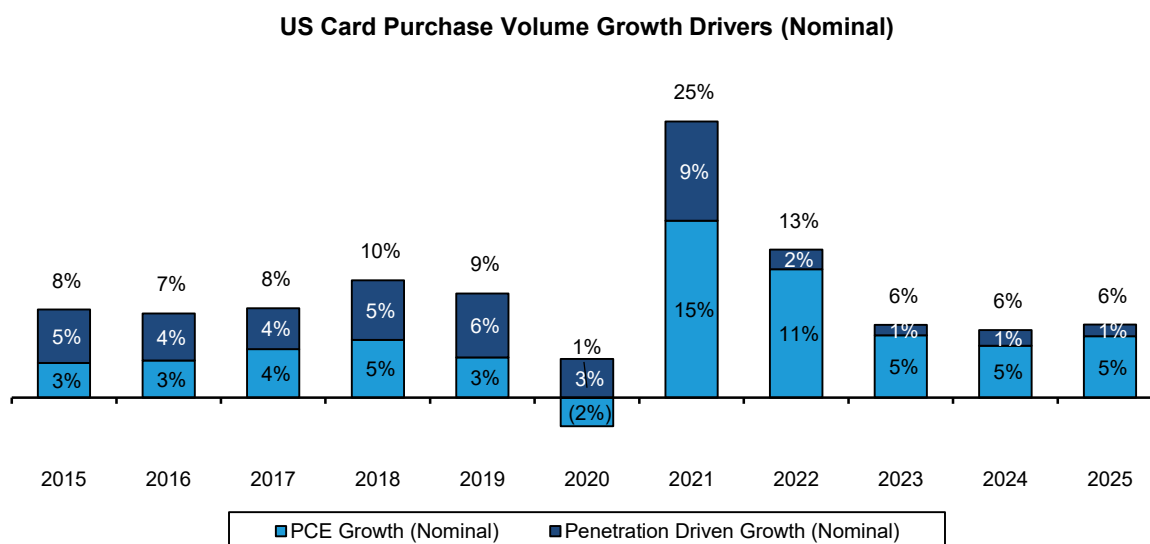
Our definition of addressable PCE (denominator) = PCE — Imputed Rentals (essentially mortgages) — Healthcare (which is not out-of-pocket) — Certain Financial Services.

For the numerator, we include Private Label Debit, Prepaid and PIN Debit and exclude Commercial volumes + our estimate of Visa direct in the US
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 15: Card Penetration is higher at 83% using a more conservative definition of PCE

Starting from Addressable PCE, we further exclude rentals, utilities and education to arrive at Adj. Addressable PCE (denominator)

For the numerator, we include Private Label Debit, Prepaid and PIN Debit and exclude Commercial volumes + our estimate of Visa direct in the US
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 16: US card volume growth has been at ~6% in the past 3 years; the growth mix has flipped with more of the growth coming from PCE growth (vs. card penetration increase historically)

Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EUROPE - A DEEPLY ATTRACTIVE MARKET FOR CARDS

With so much investor focus on the US and dynamics in EM (e.g., India and Brazil), card volume growth in continental Europe is often ignored. Europe (including UK) is ~25% of card volumes, grew mid-teens% cc CAGR between 2022-25. Ex-UK, the growth numbers are more impressive at ~20% between 2022-25. The 10 largest countries in continental Europe have >20%

of the remaining cash/check opportunity globally. Overall card (including domestic schemes) penetration in top 10 countries in continental Europe (ex-UK) is ~60%, up from ~40s% in 2016. **The networks are doing very well in Europe, not only with respect to digitizing cash, but in certain cases, taking share from domestic schemes.** In that regard, places such as continental Europe are great markets for card networks — cash is being digitized due to accelerants such as contactless and eCommerce, and unlike India/Brazil, there aren't a ton of alternative ways to pay proliferating.

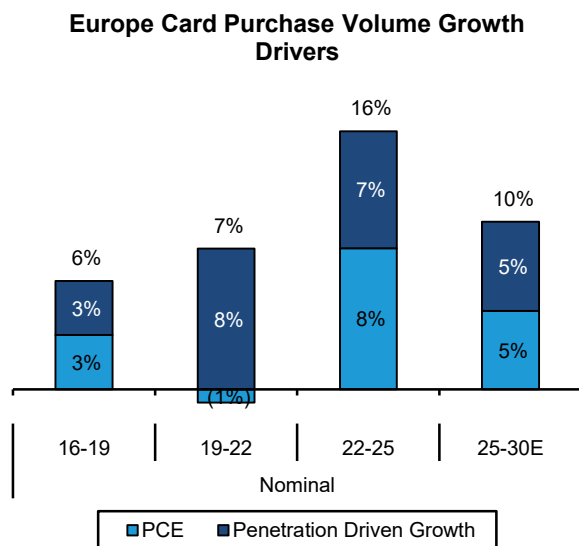
Things have changed in Europe post-pandemic (even in historically stubborn cash markets such as Germany) with a generational shift towards digital payments helped by growing contactless and eCommerce usage. Interchange caps have meant it is not expensive to accept cards. For e.g., Europe capped interchange fees at 20bps for debit and 30bps for credit in Dec 2015 - and since then card penetration has massively increased in the region (the pandemic-driven digitization also helped here). Visa Europe is no longer a not-for-profit and has drastically expanded its local presence and acceptance footprint in the continent. Fintechs have accelerated digitization and V/MA are often their partner of choice. Rising cross-border commerce has helped, since many local domestic schemes are not accepted across borders.

Card revenues are growing double digits even in markets with very high card penetrations such as the Nordics. Visa has continued to innovate in the region, and despite there being low levels of cash, has maintained its leadership in consumer payments, by partnering with digital wallets and deepening its presence in the region (Swish, Vipps) and crypto players as well as offering solutions for A2A Payments. As a result, Visa has not only managed to gain 7ppt share within card networks but has also grown overall revenue at ~15% CAGR between FY22-24 (with VAS CAGR growing 35%+ in the same time period).

Interestingly, the opportunity in Europe is not just related to cash/check. We estimate that domestic schemes drive ~\$1-2T of volumes across the top countries, and it appears that V/MA are gaining share in some instances from domestic schemes ([Exhibit 20](#)). For Mastercard, there is remaining opportunity to convert Maestro branded cards to Mastercard which is beneficial to both reported volumes and yields. See our work on Europe growth opportunity - [Payments: Europe — the secular growth opportunity](#) (March 2024)

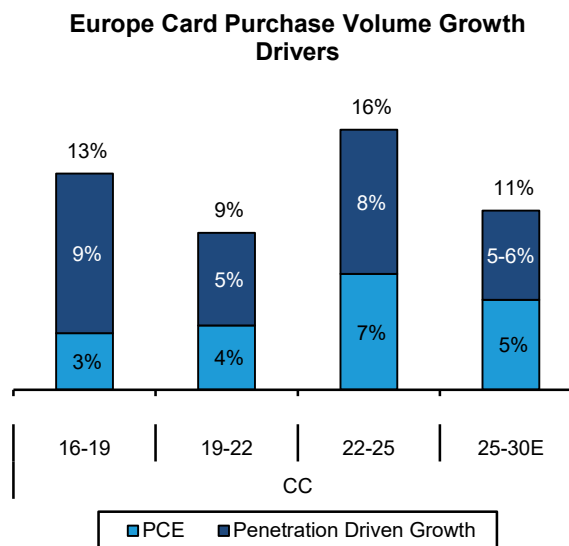
A key risk to watch in the EU is nationalism including recent initiatives from EPI (e.g., Wero). More on this in future research.

EXHIBIT 17: **From 25-30, we forecast ~10% nominal..**

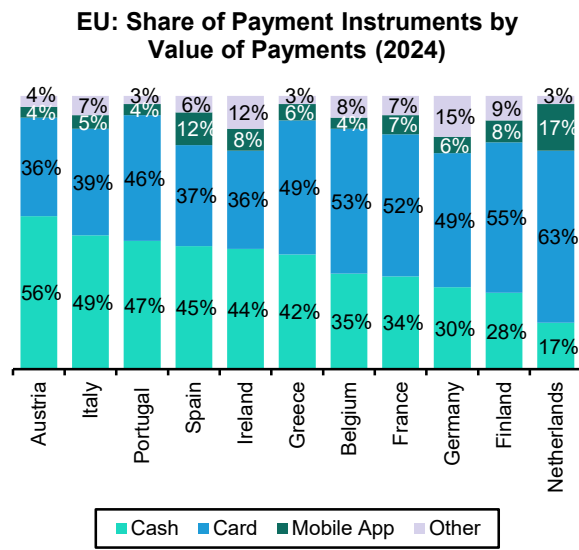


Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

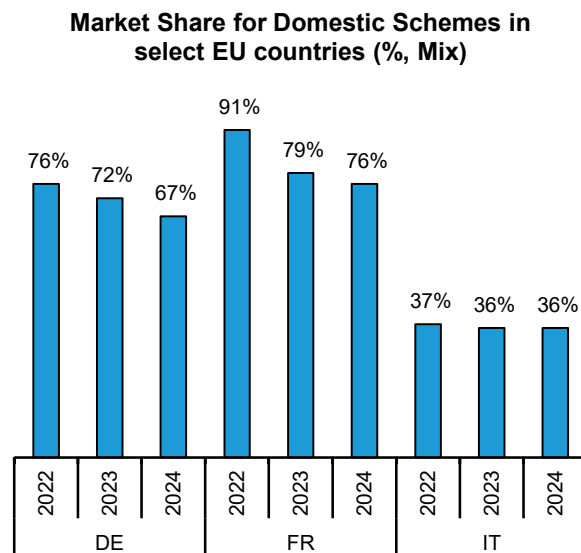
EXHIBIT 18: **..and ~11% cc volume growth in Europe**



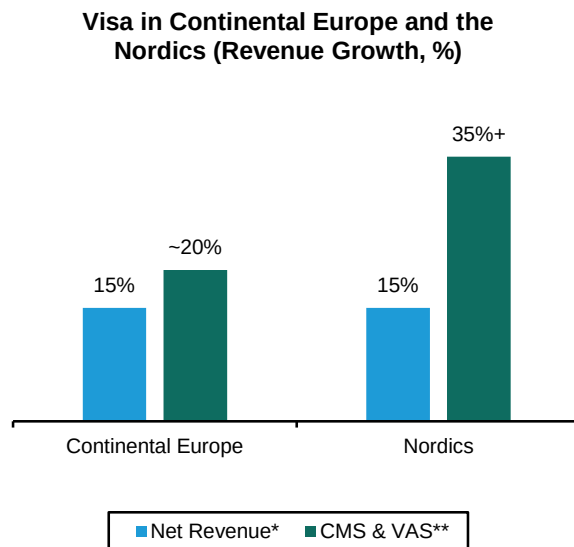
Source: Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 19: In EU, there is still heavy cash usage in many countries

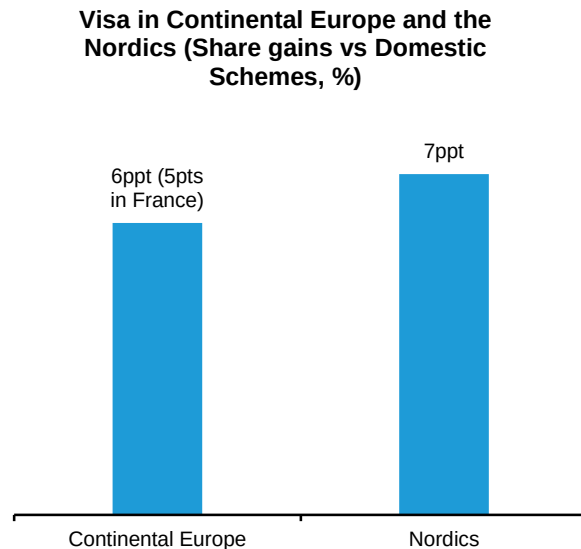
Source: ECB SPACE Study 2024, Bernstein analysis

EXHIBIT 20: Domestic schemes in select countries seem to be losing some share.. especially in Germany and France

Source: Global Payments Report, Bernstein analysis

EXHIBIT 21: Even in mature markets such as the Nordics, Visa's overall net revenue grew ~15% cc CAGR and 35% CAGR cc for CMS + VAS (2022-24)

*Net Revenue growth for Continental Europe is YoY and Nordics is CAGR 22-24

**In Continental EU, growth for both CMS and VAS was ~20% CAGR between FY22-24, in the Nordics VAS CAGR was 35%+ CAGR between FY22-24.
Source: Company reports, Bernstein analysis**EXHIBIT 22: Between CY20-23, Visa gained 7ppt share in the Nordics. Similarly in 2024, Visa has gained ~6ppt market share vs. all card players in Continental Europe**

Source: Company reports, Bernstein analysis

APAC - BATTLEGROUND REGION WITH MULTIPLE COUNTRY DYNAMICS

We are closely watching dynamics in Asia where card volume growth has been anemic due to various country-level dynamics for e.g., loss of co-badged volumes in mainland China (and negatively impacting V/MA's Asia growth numbers), poor performance of debit in India/Indonesia (where credit is currently growing DD% and represents the majority of the card spend) and digital payments' maturity in other markets such as Australia and South Korea (which has historically been dominated by domestic schemes). That said, there are developed economies such as Japan, where cash still makes up ~35-40% of overall (eComm +POS spend) today and has been a good market for cards.

That said, if we look at card penetration in the region, while penetration incl. V/MA china volumes has somewhat decelerated over the years, we note that penetration excl. V/MA China volumes continues to grow modestly ([Exhibit 24](#)).

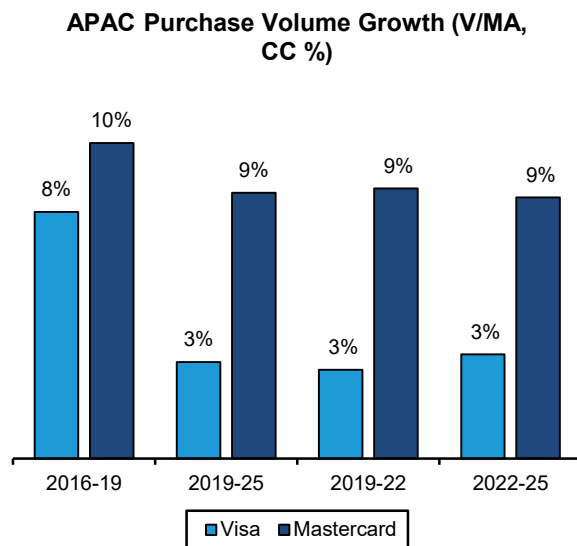
Asia also offers partnerships with local wallets which are looking for card-based monetization models and cross-border acceptance. In 2025, MA announced a new partnership with Alipay in Hong Kong and GCash to enable 36 local e-wallets to extend their use cross-border through the international Alipay+ Wallet Gateway (private). Visa has also had partnerships with local wallets in Asia. We closely watch these partnerships especially in the context of the size of these wallets in domestic markets and the fact that V/MA often don't participate in these wallet transactions, and they bring highly lucrative cross-border transactions. Digital wallets have been an area where V/MA have had good success in converting potential frenemies (e.g., digital wallets) into friends.

A note on China:

Mastercard now has a domestic processing license in mainland China (as part of a JV). We had historically believed that the ship may have sailed on the domestic China opportunity. Instead, we note some optionality now that the dynamics within the China market are changing with regulators starting to balance out the market between bank cards and digital wallets. It is also now permissible to have a card proposition that can be used domestically and in cross-border.

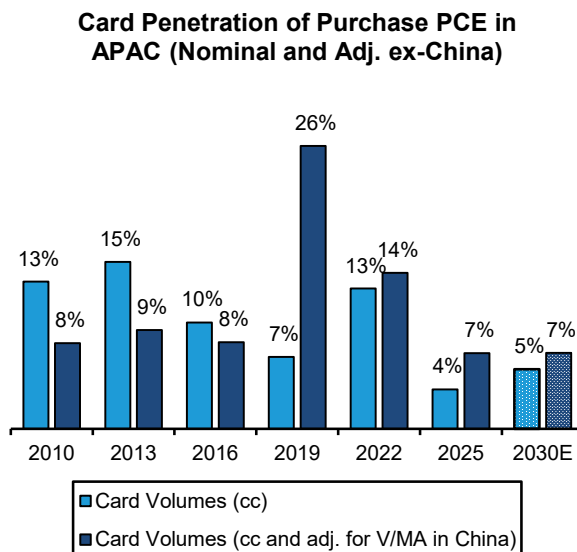
Since Mastercard has the domestic processing license (as part of a JV), it is the only party that can provide a single-use card which gives Mastercard the ability to compete (we believe it is likely in affluent centric portfolios where cross-border acceptance is a desired feature). The company is following a two-pronged approach while increasing investments in the region, by 1) Rolling out domestic switching capabilities (making it widely accepted, enabling reliable and consistent user experience for both domestic and international transactions; partnerships with ICBC, Agricultural Bank of China, and China Merchants Bank) and 2) Expanding Mastercard Move in the country (17B endpoints; partnerships with WeChat Pay, AliPay). Mastercard is working on the ground to get issuance and acceptance network which the government also wants (e.g., to have payment options for tourists). Although the mainland China opportunity could eventually be interesting, we note that it could be a multi-year process (regulatory and geopolitical risks permitting) to build a domestic processing business in China.

EXHIBIT 23: **APAC is a battleground region - where volume growth has been weak driven by unique dynamics in mainland China and some other local market nuances**



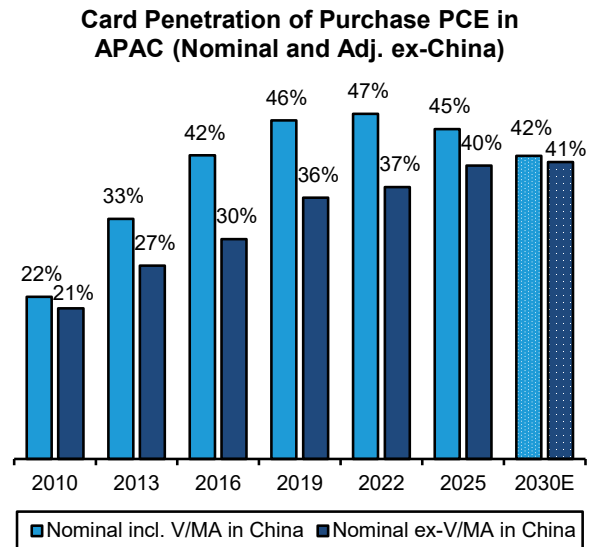
We use calendar years for Visa, and APMEA volumes for Mastercard
Source: Company reports, Bernstein analysis

EXHIBIT 25: **APAC volumes ex-V/MA in China have been growing faster vs overall volumes..**



Source: Nilson, WEO, Company reports, Bernstein estimates and analysis

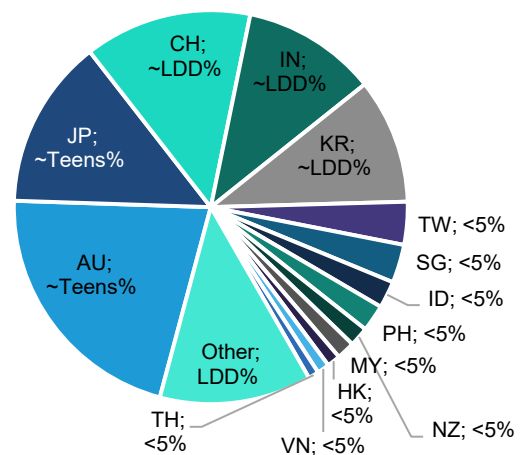
EXHIBIT 24: **While overall card penetration has been declining in the region, ex-V/MA volumes in China, card penetration has been growing modestly..**



Source: Nilson, WEO, Company reports, Bernstein estimates and analysis

EXHIBIT 26: **Our rough estimate of card volumes by market in Asia**

APAC Card Volume Mix by Market (~\$4T, 2025)

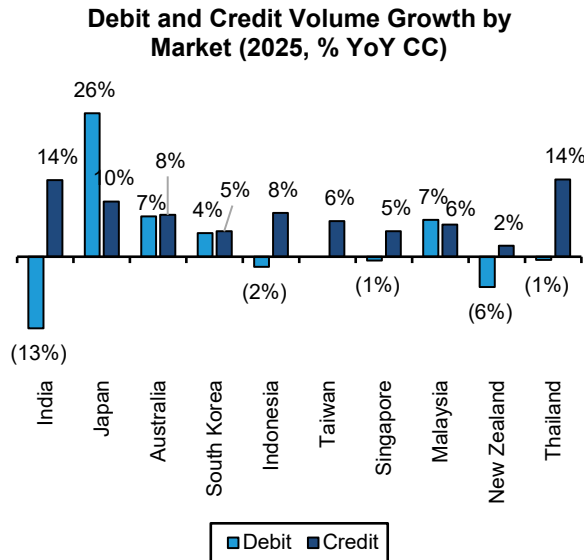


Numbers included in the exhibit are from a combination of sources i.e. Central Bank Data + Worldpay Global Payments Report. In some cases, Central Bank data may not include cards within digital wallets.

For POS + eComm 2025 numbers, we use Global Payments Report 2024 data and grow it by 2025-30 CAGR from Global Payments Report 2025

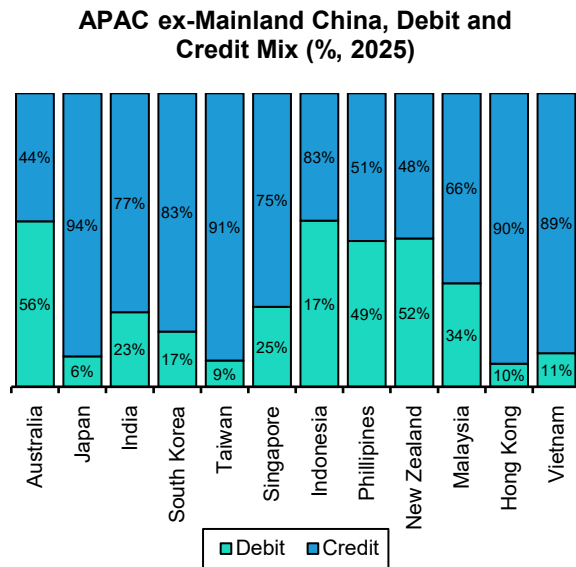
Source: Central Bank data, Global Payments Report, Bernstein analysis

EXHIBIT 27: While debit cards are performing poorly in places such as India and Indonesia, credit cards growth has been strong



Source: Central Bank Data, Bernstein analysis

EXHIBIT 28: In places such as India, credit cards account for the vast majority of volumes



Debit and Credit volumes include cards within digital wallets. Share of cards within wallets are estimated based on survey data in the WP Global Payments Report 2025

We exclude volumes from Domestic Schemes based on network market shares from the Worldpay Global Payments Report.

For Australia, we exclude EFTPOS volumes from Debit Volumes

For Japan, Korea, we exclude domestic scheme volumes from Debit and Credit volumes

For India, we exclude Rupay volumes from both Debit (assume 80% share) and Credit volumes (20% share)

For Indonesia, Taiwan, Thailand, Singapore, Hong Kong, Malaysia, Philippines, Vietnam, New Zealand, we exclude domestic schemes from Debit volumes

Source: Worldpay Global Payments Report, Bernstein analysis

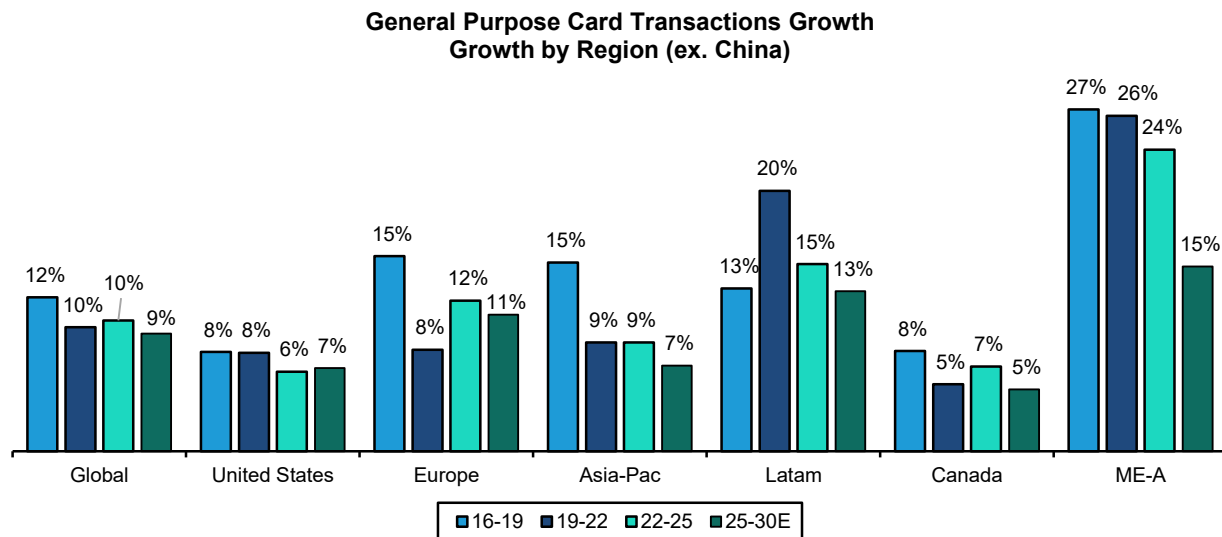
WE FORECAST 9% TRANSACTION GROWTH BETWEEN 2025-30

We forecast a stronger growth in the number of transactions at 9% growth globally between 2025-30E vs. a similar 10% in both 2019-22 and 2022-25. Card penetration is lower (and hence opportunity is greater) when assessed as a % of number of transactions (as opposed to transaction value). V/MA derive >1/3rd of their gross revenue from # of transactions, which tend to grow faster than \$ volumes. Transactions are growing faster vs purchase volume growth ([Exhibit 31](#)).

This is an important point, as processed transactions play a pivotal role in contributing to revenues for the networks. As can be seen in [Exhibit 30](#), data processing, which derives revenue on a per-transaction basis, made up 36% of Visa revenues in FY25. Further, token penetration is also growing (and will likely further grow with agentic) - and provides opportunity (including pricing) for token related value-added services. Today, ~54% of Visa's e-commerce transactions are tokenized and 100% of Agentic transactions will be tokenized.

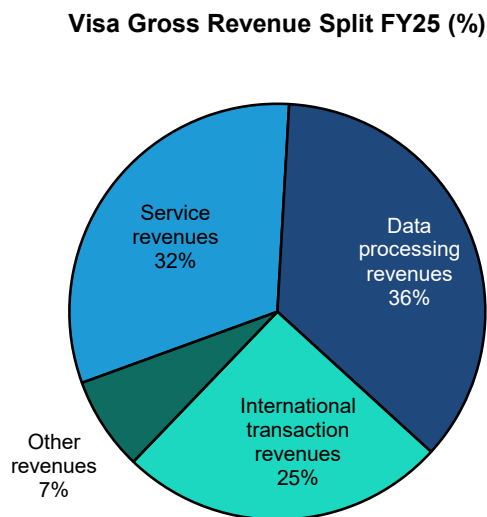
Processing penetration gains also drive incremental switched transaction growth. For e.g., Mastercard's switching penetration is now >70% (vs 60% in 2020). In addition to driving direct revenue, switched transactions also generate richer data that enables higher-margin services (fraud, analytics, loyalty) to be layered on top of the transaction, generating a flywheel effect or more switching leading to more data and services.

EXHIBIT 29: **We forecast ~9% global transactions growth, with 7% in the US, ~11% in Europe, ~7% in APAC, ~13% in LATAM, ~5% in Canada and ~15% in MEA between 2025-30E**



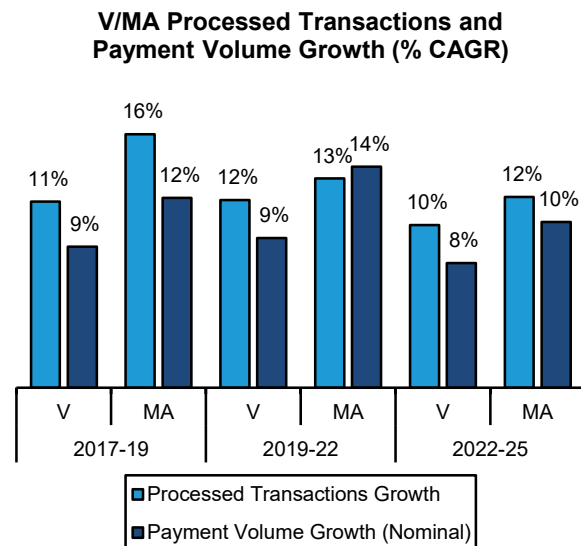
Nilson, World Bank, IMF, Bernstein estimates and analysis

EXHIBIT 30: **Transactions growth also matters along with \$ volume growth.. >1/3rd of Visa's gross revenues is assessed on a per transaction basis**



Source: Company reports, Bernstein analysis

EXHIBIT 31: **Processed transactions have historically grown faster vs. volumes**



We use calendar year for Visa

We compare 2019-25 to 2017-19 because of Visa Europe dynamics
Source: Company reports, Bernstein analysis

ALTERNATIVE WAYS TO PAY - RISKS ARE MANAGEABLE

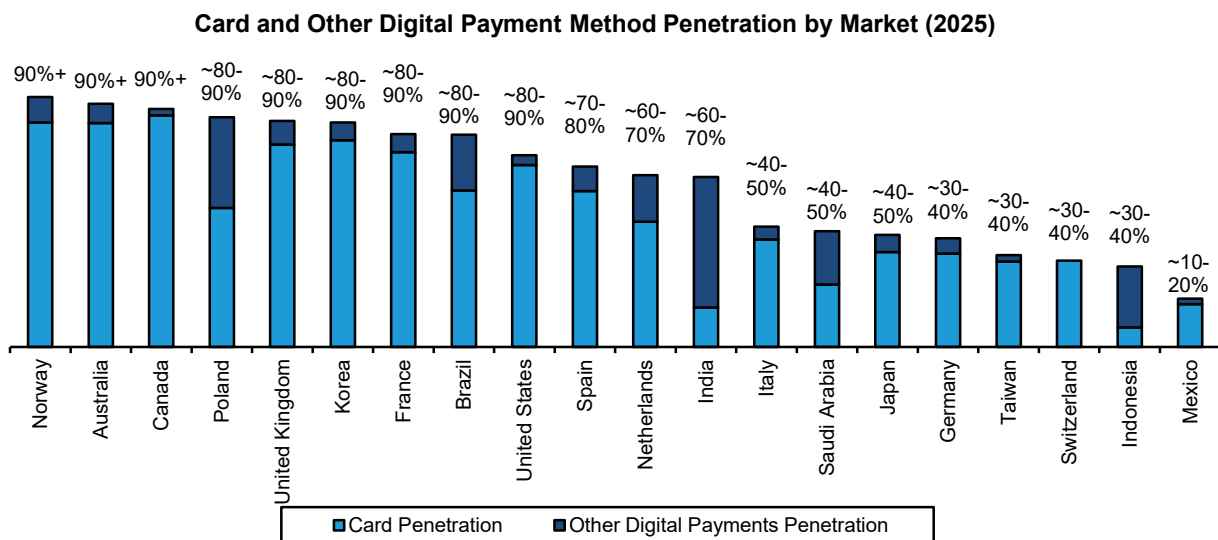
ACCOUNT-TO-ACCOUNT PAYMENTS – EXISTENTIAL RISK OR OVERBLOWN CONCERNS?

- We continue to closely watch **substitution risks from account-to-account (A2A) payments. On the surface, India and Brazil have offered powerful case studies on the potential for account-to-account or bank payments.** However, India debit market, for all the focus on this, is hardly material for the card volumes. **Credit cards payments in both India**

and Brazil are growing at HSD/double-digit rates — debit usage, however, is shrinking in India and growth is also now anemic in Brazil.

- **The real question is whether A2A payments can take share from card payments in key developed markets. Against that context, we are closely watching the evolution of pay by bank services in the U.S. and Europe.** We are watching merchant adoption for real-time payments, which has been lacking in the U.S. and Europe. We are especially monitoring recurring payments verticals such as bill pay.
- **Taking a step back, the much-feared A2A and alternative ways to pay currently represents HSD-Low-teens% of consumer-to-business payments globally.** Alternative ways to pay are not new and are less of an issue — Visa and Mastercard have co-existed with domestic schemes globally for decades. While account-to-account is a risk to watch, even in markets where A2A payments have had success (e.g., India, Brazil), credit card volumes are growing HSD/LDD%. A2A payments are a risk to debit, not credit card payments. We believe A2A limits the TAM but doesn't pose existential risks (as feared by many investors) to current volumes. In incumbent markets (e.g. US), risks from A2A are relatively contained and more around specific types of payments (e.g., bill payments, wallet funding, etc.). These categories represent a small portion of card payments. A great example is the UK, which has had RTP live for almost 15 years. In fact, after Mastercard's acquisition of Vocalink in the UK, it launched Pay-by-bank solutions in the country. Impact on Mastercard's debit volumes was minimal. The Pay-by-bank solution in the UK has hardly had traction. This is because there are still many challenges with respect to wider acceptance of A2A payments in V/ MA's core markets — especially with regard to lack of a compelling value proposition for consumers, merchants, and issuers.

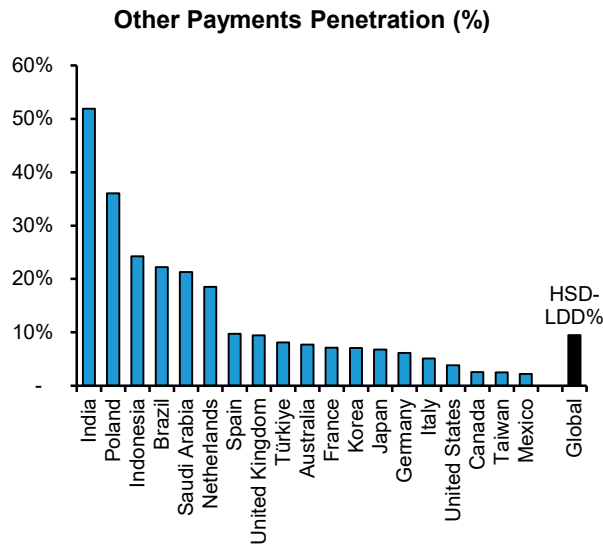
EXHIBIT 32: **Among the top 20 GDP countries, digital payments penetration varies widely..**



Ranked by respective market's overall digital payments penetration (card + other)

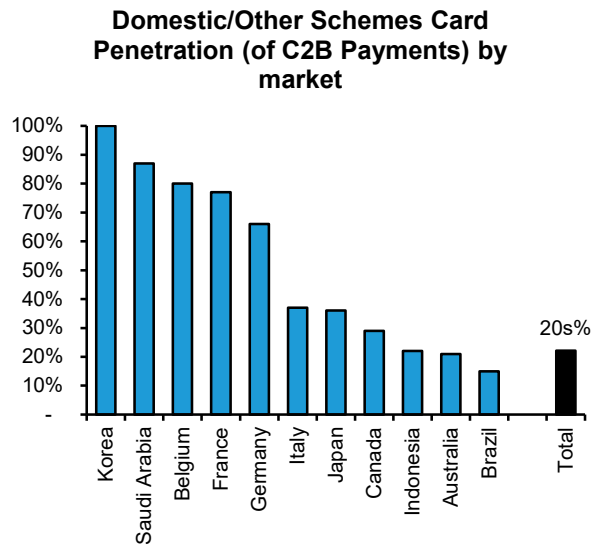
Source: Central Banks, Worldpay Global Payments Reports, IMF, Mastercard, Visa, Bernstein estimates and analysis

EXHIBIT 33: Alternative ways-to-pay drive HSD-LDD% of C2B payments globally and have had successes in places such as India and Brazil



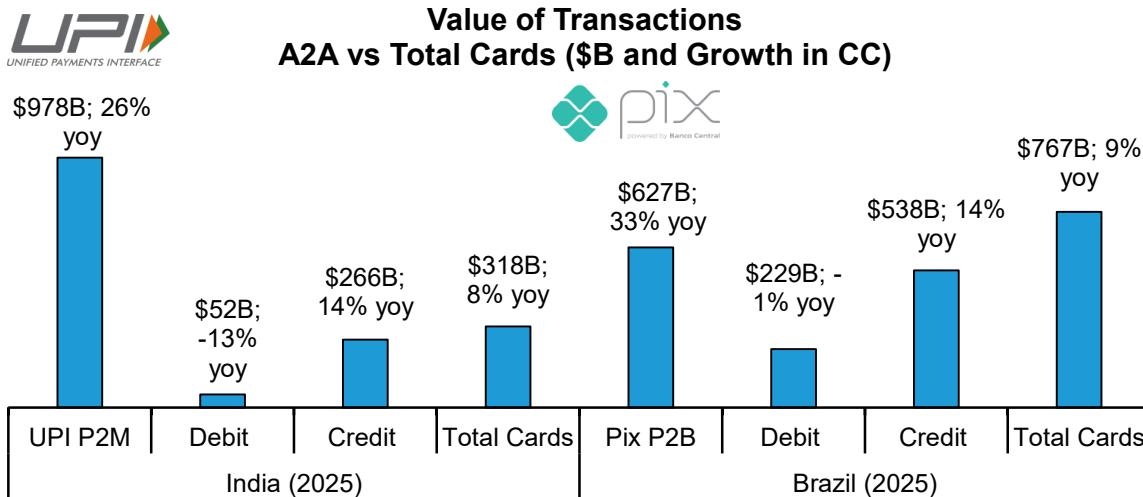
Ranked by respective market's overall digital payments penetration (card + other)
Source: Central Banks, Worldpay Global Payments Reports, IMF, Mastercard, Visa, Bernstein estimates and analysis

EXHIBIT 34: Alternative ways-to-pay are not new, domestic schemes (often co-badged with V/ MA) already drive 20s% of global card volumes



Source: Central Banks, Worldpay Global Payments Reports, IMF, Mastercard, Visa, Bernstein estimates and analysis

EXHIBIT 35: Even in countries where A2A payments have had success (e.g., India, Brazil), card volumes are still growing driven by credit



Source: Reserve Bank of India, NPCI, Central Bank of Brazil, Bernstein analysis

STABLECOINS: SOLUTION LOOKING FOR A PROBLEM IN RETAIL CONSUMER PAYMENTS

In retail C2B payments, stablecoins remain a solution looking for a problem (for the foreseeable future) in most developed markets. We haven't seen any evidence of C2B consumer payments activity happening on stablecoins in developed markets or in EM (at scale). There is the chicken-and-egg problem of building a merchant and consumer network. Debit is already regulated (& cheaper vs. credit). Cheaper rails already exist (but barely used for retail payments) e.g., ACH and RTP solutions. In a world where conversion is still a problem, we wonder to what extent merchants would want to steer customers

at the point of checkout. Offline is way harder than online. Remember, money movement (provided by different rails including stablecoins) is often a commodity, trust/governance (provided by cards) is NOT. Anyone (including payment companies) can use stablecoins for faster settlement.

Ironically, the best and one of the fastest consumer payments use-case of stablecoins is stablecoin-linked cards.

Cards often provide monetization models for many fintechs including crypto firms. Cards' value proposition (vs. other payment methods) is stronger in agentic commerce where trust becomes even more important.

Retail cross-border payments (ones addressed by cards) also remain hard to disrupt given lack of consumer friction and challenges in changing consumer behavior and acceptance infrastructure. Also, cross-border retail payments often happen in local currencies and the bulk of the fees are in the FX conversion — which dollar-denominated stablecoins also require. The challenges in off/on ramps likely drives more usage of 'stablecoin' linked cards or card infrastructure as an off-ramp for stablecoins.

Stablecoins, however, have stronger product market fit in cross-border settlement, less liquid remittance/payout corridors, cross-border B2B and treasury/cash management. All of this will take years to play out, but we are seeing increased activity, especially in markets where there is greater desirability around owning U.S. dollars (98% of stablecoins are U.S. dollar denominated).

Stablecoins may have also greater utility in the internet native agent-to-agent micro-payments (e.g., agents paying for data access, APIs). However, the space is currently very nascent (<\$50m in cumulative volumes). Finally, anyone who thinks stablecoins can somehow do micro-transactions and V/MA cannot do so, doesn't fully understand V/MA's business model (and how incentives and pricing work). Visa and Mastercard can batch-settle transactions, and can configure pricing and create commercial frameworks to make these transactions work. **We further addressed the micro-payments topic in the Agentic Commerce section below.**

Taking a step back, stablecoins offer payment companies new opportunities to make money (and NOT lose volumes), and strategically enhance their networks (e.g., through faster settlement, fiat/stablecoin conversion capabilities). There is a greater opportunity for new flows in lucrative cross-border payments. Visa and Mastercard have >150 stablecoin/crypto linked card programs. Both networks are likely monetizing stablecoins today. We are closely watching Mastercard's BVNK acquisition and the potential for greater role in orchestration and on/off ramps. It will also be interesting to see what PayPal can do with PYUSD.

See our detailed note:

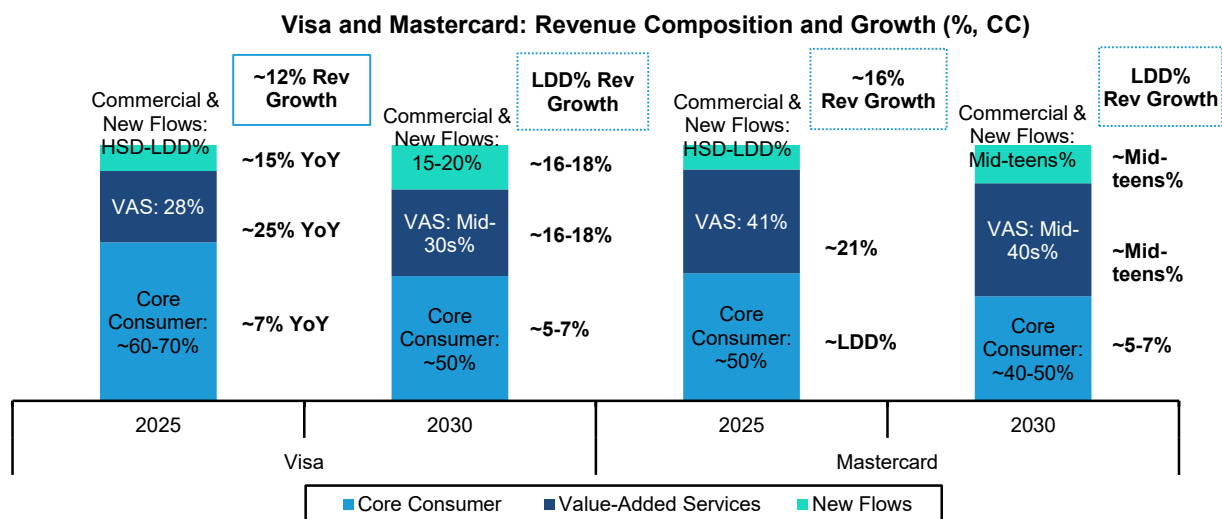
- [Payments: Stablecoins — What we learned from our dozens of industry conversations; Slide deck](#) (May 2026)
- [Stablecoins — are they a big deal?](#) (June 2025)

VAS EXTEND THE GROWTH RUNWAY; NEW FLOWS PROVIDE OPTIONALITY (MORE ON THIS IN FUTURE RESEARCH)

As shown in Exhibit 36, VAS and New Flows together represent ~40% of Visa's and ~50% of Mastercard's revenues as of calendar year 2025, with both segments compounding at 1.5-2x the core consumer growth rate and increasingly driving the growth algorithm. We expect this mix shift to extend through 2030E: Visa's VAS scales from ~28% to mid-30% of revenue with New Flows reaching 15-20%, and Mastercard's VAS expands from 41% of revenue as of 2025 to mid-40s% by 2030E. As core consumer growth normalizes toward 5-7% yoy, these higher-growth verticals should sustain group revenue growth in the LDD % range, preserving the networks' growth algorithm into the back half of the decade.

We believe new payment flows, value-added services and software will continue to expand the payments TAM meaningfully. While estimates and underlying economics vary on new flows and services, the opportunity is significant and could more than quintuple the TAM for digital payments alone. New Flows and value-added services will be critical for the networks as card growth slows as card penetration reaches 70%+. We will follow up with a more detailed note on this topic at a later date.

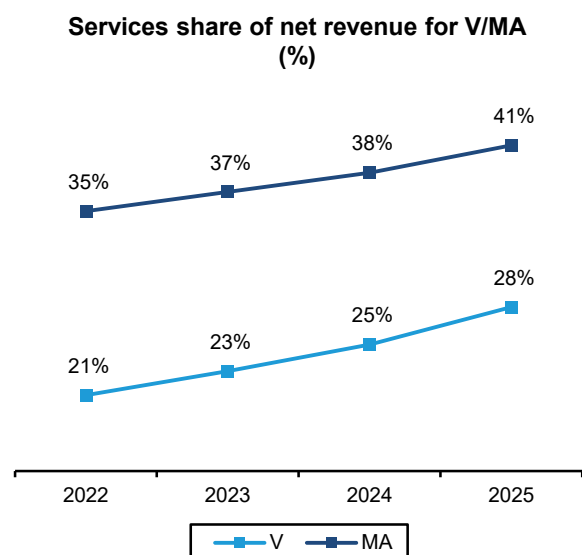
EXHIBIT 36: Value-added services and new flows, which are ~40/~50% of V/MA's revenues and growing 1.5-2x the rate of core growth, will be incremental to growth. VAS and new flows can sustain MA's revenue growth in the LDD range even in 5 years as card volume growth incrementally matures



On Calendar Year

Source: Company reports, Bernstein estimates and analysis

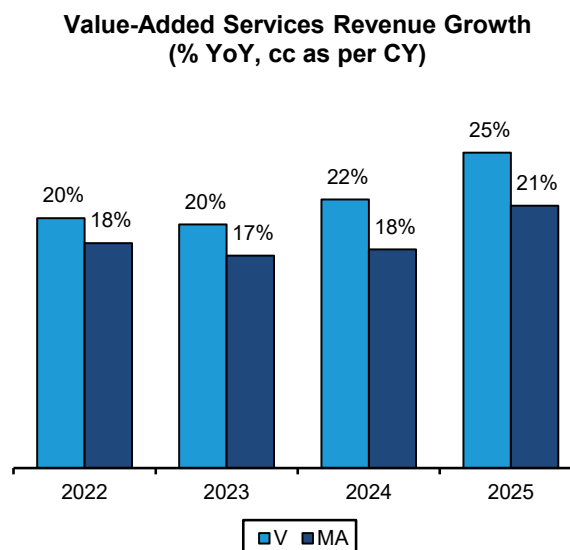
EXHIBIT 37: Value-added services make up ~40% of Mastercard's and ~30% of Visa's net revenue



Visa using calendar year

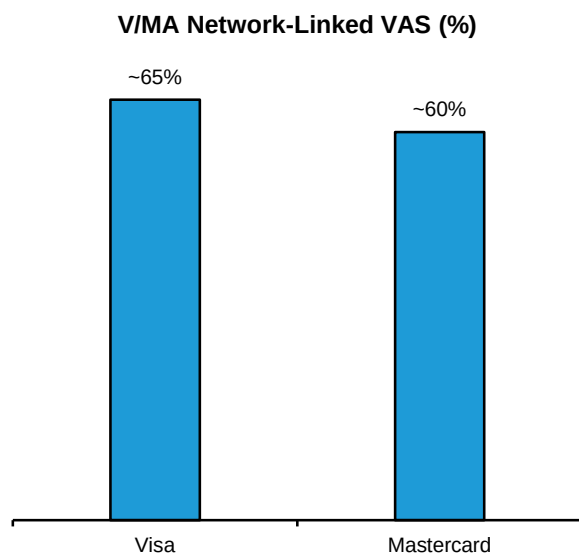
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 38: Value-Added Services grew 25% for Visa and 21% for Mastercard in CY25



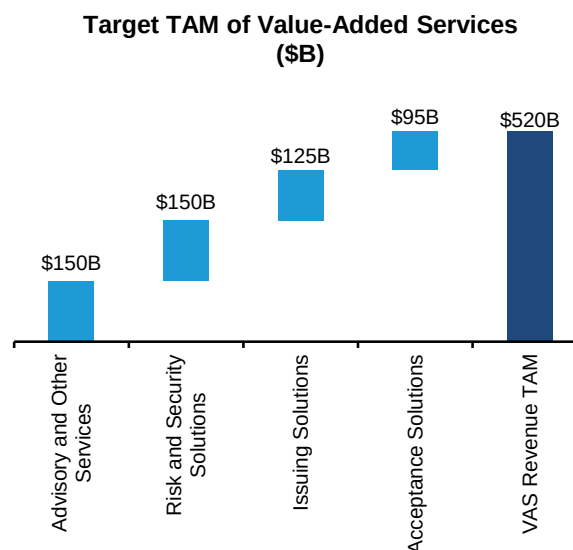
Source: Company reports, Bernstein analysis

EXHIBIT 39: ~65% if Visa and ~60% of Mastercard's VAS is network-linked; transactions are the primary driver of these services



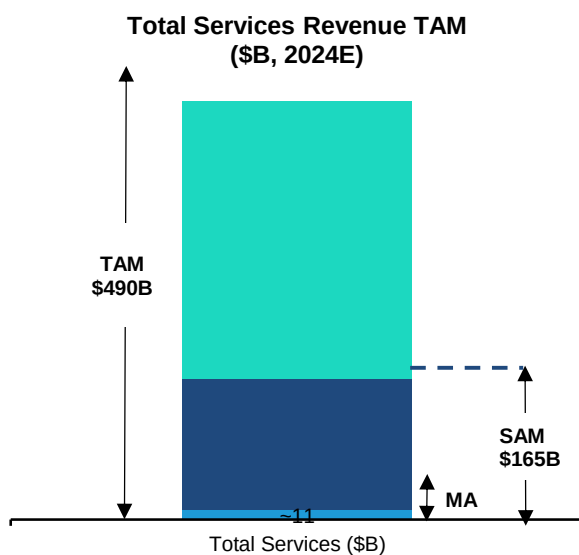
Source: Company reports, Bernstein analysis

EXHIBIT 40: During its Investor Day, Visa broke down different components and sized the revenue TAM for Value-Added Services at \$520B



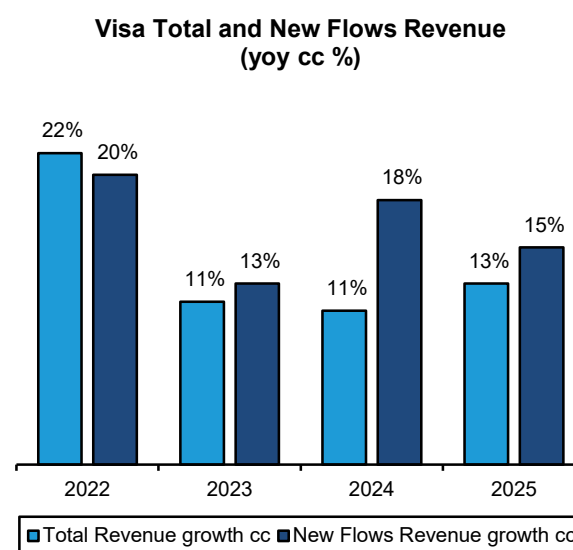
Source: Company reports, Bernstein analysis

EXHIBIT 41: Mastercard also provided a similar TAM sizing on Services Revenue at \$490B, with \$165B in SAM



Source: Investor Day Presentation, Bernstein analysis

EXHIBIT 42: New flows revenues is likely growing faster vs. the core. Note: Visa numbers are for calendar year



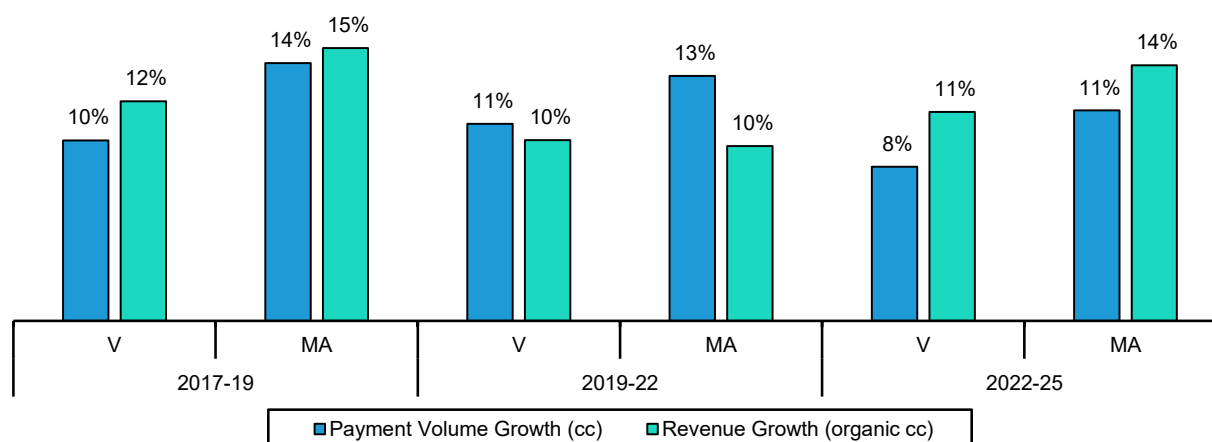
Visa using calendar year

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 43: Growth of value-added services is one of the reasons why V and MA have sustained DD revenue growth outpacing the volume growth in recent years.

Note: 2025 revenue growth likely benefited 1-2ppt from FX vol

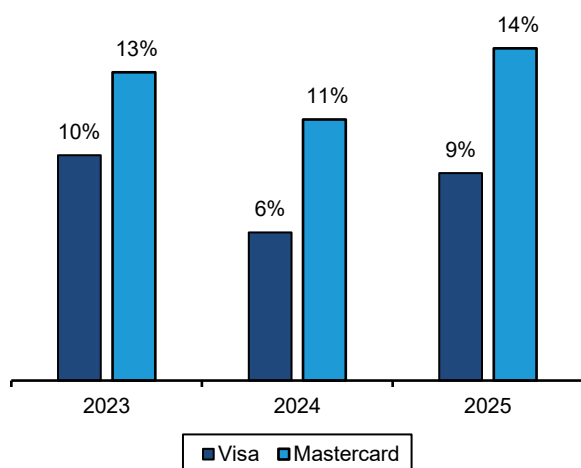
Overall Revenue (Organic CC) vs Volume Growth (CC, as per calendar years)



We use calendar quarters for Visa; we use 2017-19 to account for the dynamic around Visa Europe
Source: Company reports, Bernstein analysis

EXHIBIT 44: Both Visa and Mastercard's commercial volume growth accelerated during the last year

Visa, Mastercard Commercial Volume (Calendar Year, % YoY CC)



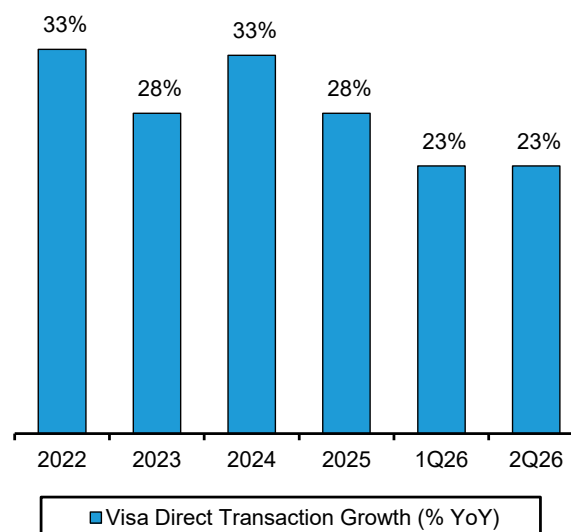
Both reflect calendar year, cc growth;

Visa's commercial growth includes large, medium and small business credit and debit, as well as commercial prepaid volume;

Mastercard commercial growth reflects MA's Commercial Gross Dollar Volume (GDV) growth, which represents purchase volume plus cash volume
Source: Company reports, Bernstein analysis

EXHIBIT 45: Visa Direct Transactions Growth (% YoY, Fiscal Year)

Visa Direct Transaction Growth (% YoY)



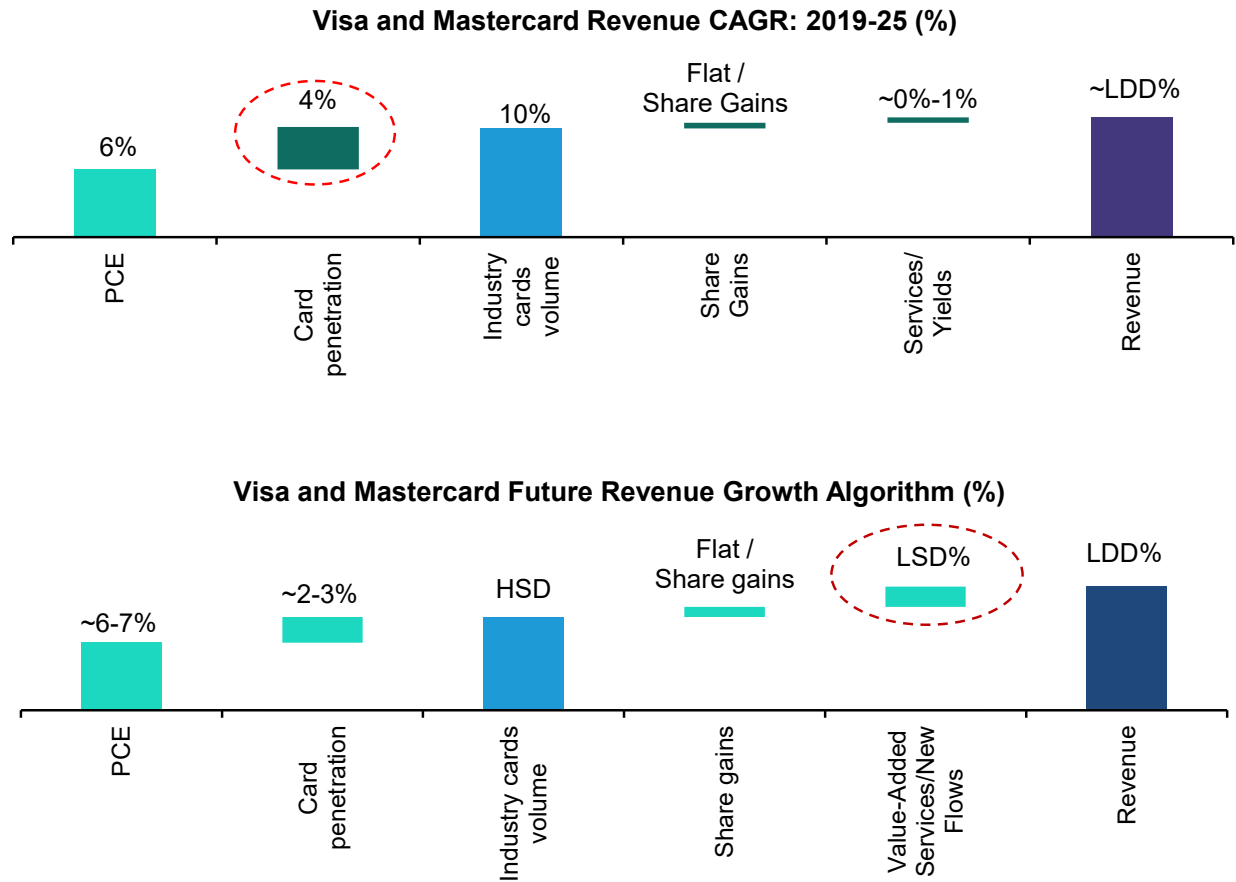
On Visa Fiscal Year

Source: Company reports, Bernstein analysis

CONCLUSION: NETWORKS' CHANGING GROWTH ALGORITHM

We expect Visa and Mastercard's growth algorithm to change as new flows and services become a larger share of the revenue mix and grow faster than core Consumer-to-Business payments (Exhibit 46). We do believe that expansion of TAM into new flows, VAS and overall diversification (at healthy margins) is a key theme to watch over the coming years, as traction here will influence the exit multiples for the networks in 3-5 years when consumer payments start maturing globally.

EXHIBIT 46: We expect Visa and Mastercard's revenue growth algorithm to evolve from card penetration growth being the #1 growth driver to new flows and services combined being the dominant growth drivers in the future



Source: Company reports, Bernstein estimates and analysis

BERNSTEIN TICKER TABLE

Ticker	Rating	8 Jun 2026		Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
ADYEY (Adyen)	O	USD	9.52	18.63	(74.9)%	USD	0.39	0.50	0.60	24.3	18.9	15.9
ADYEN.NA (Adyen)	O	EUR	833.30	1,600.00	(64.2)%	EUR	33.62	42.83	51.05	24.8	19.5	16.3
XYZ (Block Inc)	O	USD	69.92	85.00	(15.8)%	USD	2,084	3,347	4,700	19.9	12.4	8.8
FIS (FIS)	M	USD	39.54	73.00	(74.7)%	USD	5.76	6.30	6.91	6.9	6.3	5.7
FI (Fiserv)	M	USD	52.72	76.00	(91.8)%	USD	8.64	8.06	8.76	6.1	6.5	6.0
GPN (Global Payments)	M	USD	64.50	86.00	(39.5)%	USD	12.22	13.89	16.13	5.3	4.6	4.0
KLAR (Klarna)	M	USD	16.55	20.00	NA	USD	65.1	305.3	461.7	44.6	9.5	6.3
MA (Mastercard)	O	USD	485.67	710.00	(41.1)%	USD	17.01	19.82	22.82	28.6	24.5	21.3
PYPL (PayPal)	M	USD	41.26	45.00	(67.2)%	USD	5.31	5.23	5.30	7.8	7.9	7.8
TOST (Toast)	O	USD	24.10	39.00	(68.7)%	USD	633.00	802.78	990.80	19.3	15.2	12.3
V (Visa)	O	USD	319.67	450.00	(37.1)%	USD	11.47	13.17	15.16	27.9	24.3	21.1
SPX			7,405.73									
EDME			1,542.17									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

XYZ estimate is EBIT (M); KLAR estimate is Adjusted EBIT; TOST estimate is EBITDA (M); XYZ valuation is EV/EBIT (x); KLAR valuation is EV/Adj EBIT (x);

TOST valuation is EV/EBITDA (x);

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

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Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

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The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

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The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

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Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.1%	16.5%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	36.3%	17.8%
Underperform	SELL	12.6%	14.9%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of March 31, 2026. All figures are updated quarterly.

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